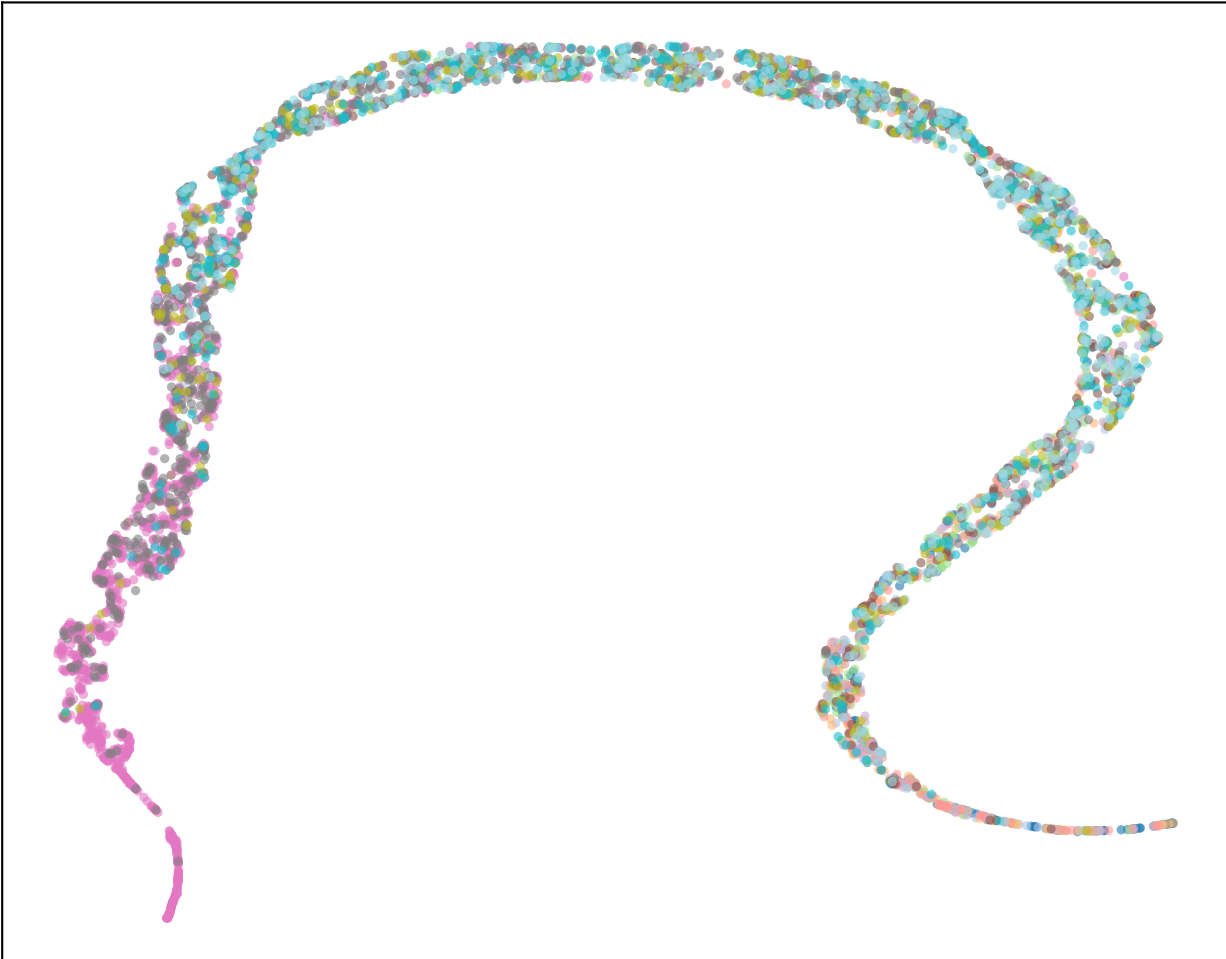


UMAP of GAP Embeddings: Untrained vs Trained
(colored by acoustic similarity clusters from raw spectrograms)

Untrained (random weights)



Trained



Acoustic similarity clusters (hierarchical clustering on raw spectrograms):
C0: DOCUMENTS, DOWNLOADS, FAVORITES, ITALICS, MINIMIZE, MMS, ... (9 total)
C1: CALCULATOR, SATURDAY, SEPTEMBER, SEVENTEEN, SIXTEEN, SUBJECT
C2: APPOINTMENT, DOCUMENT, FEBRUARY, HIBERNATE, HOMEPAGE, INBOX, ... (10 total)
C3: CALENDAR, EXPORT, FACEBOOK, FIFTEEN, FORWARD, FOURTEEN, ... (16 total)
C4: ATTACHMENT, DECEMBER, INSERT, JANUARY, NINETEEN, NOVEMBER, ... (10 total)
C5: BROWSER, CANCEL, MESSAGE, SISTER, SIXTY, THURSDAY, ... (8 total)
C6: ATTACH, AUGUST, CLOSE, FLASH, LAST, LEFT, ... (14 total)
C7: ALL, CLICK, DOUBLE, EIGHT, FOUR, FULL, ... (21 total)
C8: ADD, AS, BACK, BOLD, CALL, CHECK, ... (40 total)
C9: FIFTY, SAVE, SCREEN, SELECT, SEND, SET, ... (11 total)
C10: BEGIN, BROTHER, CAMERA, E-MAIL, FILE, FIVE, ... (15 total)
C11: A.M., ALARM, APRIL, CHANGE, COMPUTER, DAUGHTER, ... (21 total)